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1. Business analytics	- Transform data into useful insights for business decision - uses predictive, descriptive and diagnostic analysis - used to improve efficiency - used to increase revenue - used to reduce cost	Using sales data to decide what products to discontinue	Investopedia
2. Data analytics	- using data to identify actionable insights to make decisions - involves collecting, cleaning, analyzing and visualization of data	analyzing customer transaction data to determine buying trends	Mckensay
3. Business Intelligence	reporting of past/historical business performance	a power BI dashboard showing past sales data	microsoft BI
4. Data-Driven Decision making	- using data to make decisions - facilitated by data analyst	Choosing a market strategy based on past results	Harvard Business Review
5. Big data	- Extremely large datasets - collected from many sources - typically cannot be managed or analyzed using traditional data processing tools - complex datasets - structure, unstructured, semi-structured data	data from social media	IBM
6. Descriptive descriptive analytics	- used by business analyst - used to summarize historical data to understand trends and performance	monthly sales report	Investopedia
7. Diagnostic analytics	- used by business analyst - used to understand the 'why' in the data to identify cause and correlation	analyzing monthly sales report to determine why sales dropped	Mckensay
8. Predictive analytics	- used by business analyst - use statistical models - use machine learning - Goal to forecast future outcomes using data	predicting/predicting next months sales	IMP
9. Prescriptive analytics	- used by business analyst - uses optimization techniques to recommend actions to achieve desired outcome	recommendations to lower unit price of certain products to increase sale volumes	Gartner
10. Data literacy	- competency to use data - competency to read, understand, analyze and communicate with data	Reading a chart	MBR

Foundational Concepts of Analytics

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11. Structured Data	- Data in a table - Data in rows and column - Data with a consistent format	Excel spreadsheet	IBM
12. Unstructured Data	- Data not in a consistent format - Data not in a table format	Emails	IBM
13. Data Warehouse	- Stores big data - Stores structured data - House data from multiple databases - Located centrally (Eg head office)	Shoprite database for all sales data	Amazon AWS
14. Data Mining	- Analyzing large datasets to find patterns/correlations - Use statistical analysis, machine learning and computer techniques - Works on structured and unstructured	Analyzing to find buying habits of customers	Investopedia
15. Data Visualization	- Used by data analysts - Final step in the data analysis process - Showing the data in charts and graphs	Bar charts of sales over the last twelve months	Tableau
16. Dashboard	- Is data visualization in real time - Contains key metrics - A summary of key metrics	Weather forecast dash board	Google Analytics
17. Metric	- A measure / measurement - Use to quantify something	No. of transaction	Hubspot
18. KPI Key Performance Indicator	- Key metrics - Key measures / measurement - Quantifiable	Sales revenue target	Hubspot.
19. Benchmark	- Baseling - Initial point of reference - Used for comparison	Comparing company sales to industry average	Statista
20. Segmentation	- Grouping - Grouping into distinct category base on common shared characteristics	Grouping by age	Hubspot.

Data , Measurements & Reporting

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21.	- Metric - unit percentage - defined as % of people who take action	10 out of 100 customer buy Conversion rate = 10%	Google Analytics
22.	- Metric - cost to get one customer	Spend £1000 to get 10 customers = £100 each	Investopedia
23.	- Metric - Total money a customer brings over time	A customer spends £6000 over 2 years	Hubspot
24.	- Metric - unit percentage - defined as % of customers who leave	10 out of 100 customers leave Churn rate = 10%	Google Analytics
25.	- Metric - unit percentage - defined as % of customers who stay	90 out of 100 stay Customer Retention Rate = 90%	Google Analytics
26.	- Metric - measures if customers recommend you	Survey asking 'Would you recommend us?'	Bain & Company
27.	- Metric - Average money per purchase	$\frac{\text{Total sales}}{\text{Number of orders}}$	Shopify
28.	shows process of customer from initial steps awareness to buying product/service	Seed Ad ↓ Visit shop ↓ buy	Hubspot

Customer & Sales KPI

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29. Marketing Campaign	- Communication tool for - Communicates to large audience - Communicate specific messages - Planned - Runs for specific period	Black Friday specials	Google
30. Post-Campaign Analysis	analysing campaign results after campaign ends	Reviewing sales after an ad campaign	Google
31. Incremental Revenue	- additional Revenue as a direct result of a campaign / specific initiative - metric	Increased long term sales because of ads	Google
32. ROI (Return on Investment)	- metric - Profit compared to cost	Spend £1000 Earn £2000 $ROI = \frac{2000}{1000} = 2$ good	Investopedia
33. ROAS (Return on Ad Spend)	- metric - defined as Revenue per ad cost	£5 earned for every £1 spent	Google Ads
34. Click-Through Rate (CTR)	- metric - unit percentage - defined as % of people who click on ad	Clicks from 100 views = 5%	Google Ads
35. Bounce Rate	- metric - unit percentage - define as % of visitors who leave quickly	users leave website after one page	Google Analytics
36. A/B Testing	- Alpha - beta test to compare two versions to see which is better	Two ads with different headlines	HubSpot
37. Lead Generation	getting potential customers	collecting emails from a website form	HubSpot
38. Marketing Attribution	Finding which channel cause of sale	sales came from Facebook Ads	Google Analytics

Marketing & Campaign Analytics

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